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Refining Bird Species Identification through GAN-Enhanced Data Augmentation and Deep Learning Models

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Abstract

This work addresses the challenge of classifying visually similar bird species, a task complicated by subtle interspecies variations. We focused on ten bird species, assembling a dataset of approximately 8000 images from Google Images. These species were specifically chosen for their high degree of similarity, presenting a unique challenge for classification algorithms. To enhance our dataset and improve classification accuracy, we employed Generative adversarial networks (GANs), a state-of-the-art generative adversarial network, to augment our original dataset with synthetic yet realistic images. This augmentation aimed to provide a more prosperous, diverse training environment for our deep learning model. Subsequently, we developed a specialized multi-classification model tailored to recognize and differentiate these closely related bird species. Integrating GANs like StyleGAN3-augmented data into our training process represents a novel approach to ecological image analysis, potentially setting a new standard for accuracy and efficiency in classifying highly similar species. This study demonstrates the effectiveness of advanced generative models in complex classification tasks and contributes a valuable methodology to ecological research and species identification.

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1. Introduction

Accurately classifying bird species from images is crucial in ecological research and biodiversity conservation. This process is essential for tracking and monitoring avian biodiversity and plays a key role in understanding ecological interactions and shaping conservation policies [1]. However, classifying species with high visual similarity

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poses significant challenges due to nuanced differences in plumage, size, behavior, and morphological characteristics, especially against the diverse backgrounds of natural habitats [2, 3].

The challenge is intensified by limitations in existing image datasets, which often lack the diversity and volume needed to effectively train robust classification models, particularly for species with subtle differences. Traditional image classification methods frequently fall short in these fine-grained scenarios [4, 5, 6].

This work aims to develop an innovative multi-classification model in high-similarity labels with subtle variations. To address the limitations of this kind of dataset, we determined to collect the data manually and image by image, yet we faced several challenges that made the dataset limited and scarce. As a result, we employed GANs like StyleGAN3 to generate and augment the data collected earlier manually to increase the number of images in our dataset and achieve efficient performance [7, 8]. Our key contributions are summarized as follows:

- We compiled a dataset of 8000 images for ten visually similar bird species and used StyleGAN3 to generate synthetic images, enriching the dataset.
- We developed a deep learning model for accurate multi-class bird species classification and integrated GANs for ecological image analysis, achieving 93% accuracy with GAN-augmented data.

Our study explores the use of GAN for data augmentation in ecological image analysis, a significant advancement in the field. Along with developing a specialized deep learning model, this approach addresses a critical gap in ecological studies and sets a new precedent for future research [9, 10]. By demonstrating the effectiveness of this method, our research contributes new insights and methodologies to machine learning and biodiversity conservation [11, 12].

The rest of this paper is structured as follows. In Section 2, we review relevant literature. Section 3 describes our methodology, including data collection, preparation, and deep learning model development. Section 4 presents our results and discussions. Finally, Section 5 concludes the paper and outlines future research directions. The Acknowledgments and References follow.

2. Literature Review

Accurate bird species classification is crucial for ecological research and biodiversity conservation. It aids in monitoring avian biodiversity, understanding ecosystem dynamics, and shaping conservation strategies. Bird migration patterns can also signal climate change effects. However, species identification is challenging, especially for visually similar species [5, 13].

On the one hand, traditional classification methods, based on field observations and manual techniques, suffer from scalability issues and observer subjectivity [14, 15]. The advent of automated classification, while addressing these limitations, faces difficulties distinguishing subtle differences in appearance among similar species. Factors like lighting and movement further complicate automated systems [16, 17, 18].

On the other hand, image classification has evolved from basic feature extraction to advanced deep learning models, particularly convolutional neural networks (CNNs). These models handle natural image variability, significantly improving species classification [19, 10, 20, 21].

Furthermore, machine learning model efficacy depends on diverse, high-quality training data. However, gathering such data, especially for rare or visually similar species, is challenging [22, 23]. Data augmentation, including geometric and color space transformations, partially addresses this but lacks uniqueness [24, 25]. Additionally, Generative Adversarial Networks (GANs), specifically StyleGAN and StyleGAN3, revolutionize data augmentation by generating varied, realistic synthetic images. This enhances dataset diversity, benefiting models trained on limited real-world data [7, 8, 26, 27]. Though emerging, GANs' application in ecological image analysis faces challenges in ensuring the ecological accuracy of synthetic images [28, 29, 23].

In conclusion, while advanced deep learning models have significantly advanced bird species classification, their success hinges on dataset diversity. GANs like StyleGAN3 offer promising solutions for data augmentation, but their use requires careful validation to maintain ecological accuracy.

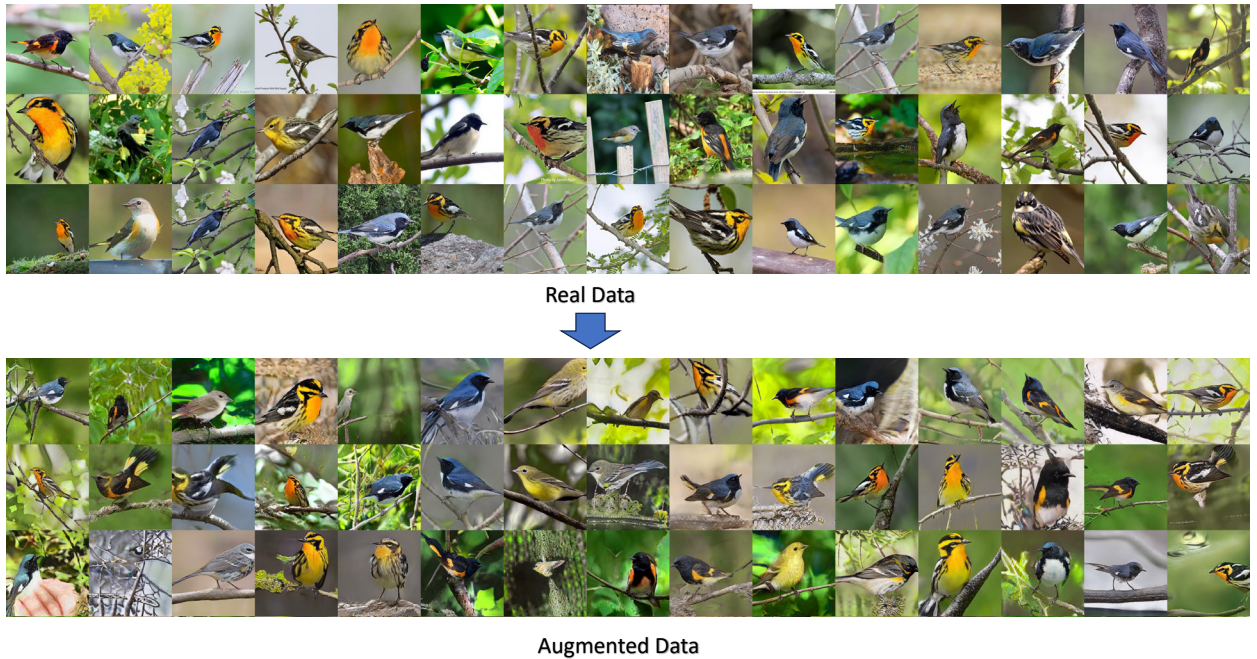


Fig. 1. Comparison of Original and GANs Augmented Bird Images - The top panel showcases a diverse array of original bird images from our dataset, capturing the species in their natural habitats. Below, the bottom panel demonstrates the augmented images produced by StyleGAN3, highlighting the model's adeptness at replicating the complex visual details characteristic of each species.

3. methodology

3.1. Dataset Collection and Preparation

In this comprehensive study, we embarked on a meticulous journey to source and prepare a dataset specifically tailored for classifying bird species that exhibit close visual similarities. Recognizing the challenges inherent in distinguishing these species, we employed a rigorous approach to data collection, processing, and labeling. Our objective was to build a rich and diverse dataset that would not only facilitate accurate species classification but also provide insights into the nuances of avian biodiversity. The following subsections detail our strategic methods in selecting bird species, collecting and processing images, and preparing the dataset to develop a robust and efficient classification model.

3.1.1. Data sourcing

Selection of bird species. We focused on 10 bird species known for their visual similarities to each other, posing a unique challenge for classification. The species were selected based on specific criteria, such as the subtlety of inter-species differences and their prevalence in certain ecological studies.

Image collection process. Approximately 8000 images were sourced from Google Images. A custom script was developed to unify the image extension to “.jpg,” using carefully crafted search queries to target images that feature each bird species prominently. While manually collecting the images, our observation filtered them based on resolution and clarity to ensure high-quality data. Additionally, we tried to select different positions and genders for the species during its life cycle.

Diversity and representativeness. Special attention was given to the diversity of the images. Each species' dataset included various poses, genders, baby and premature birds of the same species, environmental backgrounds, and

lighting conditions. This variety is critical to ensure that the models trained on this dataset can generalize well across different real-world scenarios.

3.1.2. Image processing and labeling

Preprocessing steps. All collected images were preprocessed to conform to a standard format suitable for training. This included resizing images to a uniform dimension (e.g., 224x224 pixels), converting them to a consistent color space (RGB), and standardizing the file format (JPEG).

Quality assurance. Each image was manually inspected to confirm the correct identification of the species and the image's suitability for training purposes. This step ensured the removal of mislabeled, low-quality, or irrelevant images from the dataset.

Dataset labeling. The images were labeled with their corresponding species names. Each image was assigned a categorical label, representing one of the 10 bird species. This labeling process was crucial for supervised learning in the subsequent classification model training.

Data augmentation techniques. In addition to the StyleGAN3-generated images, traditional data augmentation techniques were applied to further enhance the dataset's diversity. These techniques included random rotations, horizontal flipping, and slight color variations. Such augmentations help make the model robust to new, unseen data variations.

Dataset splitting. The final dataset was split into three sets: training, validation, and testing. The training set comprised 70% of the images, the validation set 15%, and the test set the remaining 15%. This separation was essential for training and objectively evaluating the performance of the classification models.

3.2. StyleGAN3 Training for Data Augmentation

Traditional data augmentation is widely used in various machine learning models, including image classification, object detection, and segmentation in supervised learning where labeled data is available. However, traditional augmentations, such as rotation, flips, crop resizing, and zooming, are insufficient to achieve high performance. Generative adversarial networks can operate with unsupervised learning and don't necessarily require labeled data, making it a state-of-the-art data augmentation.

To augment the dataset for bird species classification, we chose StyleGAN3 for its advanced capabilities in generating high-quality, realistic images. This choice was pivotal, given StyleGAN3's proficiency in synthesizing detailed and diverse images, which is essential for capturing the subtle nuances of various bird species.

Hardware and software synergy: Our computational setup was centered around two high-performance GPUs: the Tesla V100-PCIE-32GB and the Tesla V100S-PCIE-32GB. These GPUs, known for their robust performance, were crucial in handling the demanding requirements of StyleGAN3's training process. Operating under NVIDIA driver version 530.30.02 and CUDA version 12.1, our system was well-equipped to manage intensive computational tasks. The software framework for this endeavor was built on PyTorch, a choice motivated by its flexibility and dynamic computational graph, making it highly suited for deep learning applications.

Preparing for the GAN: Prior to training StyleGAN3, a meticulous preprocessing of our dataset was undertaken. This included resizing images to a uniform resolution and normalizing pixel values, aligning them with the input specifications of StyleGAN3. Ensuring consistency in the dataset was paramount for the effectiveness of the subsequent training.

Navigating the hyperparameters: The training of StyleGAN3 was a delicate process, hinging on the careful selection of hyperparameters. We calibrated the learning rate to balance swift convergence and training stability. The batch size was optimized, taking into account the memory capabilities of our Tesla V100 GPUs, to maximize the efficiency of the training.

The training process: Spanning seven days in December 2023, the training process was a testament to patience and precision. We closely monitored the interplay between the generative and discriminative components of the network, ensuring a steady progression toward convergence. Regular checkpoints were a crucial part of our strategy, enabling us to capture the development of the model's capabilities and providing a safety net against interruptions.

Synthesizing a new reality: Upon the completion of training, StyleGAN3 embarked on its crucial task: generating synthetic images. By exploring diverse points in the model's latent space, the network produced various images, each depicting our selected bird species in different poses and settings. These images were not just mere replicas; they were new creations that expanded the breadth of our dataset.

Fig. 1 illustrates the outcome of this process, showcasing the transformation from real images to their synthetic counterparts. The top panel displays a selection of original bird images from our dataset, while the bottom panel features the corresponding synthetic images produced by StyleGAN3. This visual representation highlights the model's ability to replicate intricate details such as feather textures and naturalistic backgrounds, confirming the success of the augmentation process.

Merging the augmented data with the real: The integration of these synthetic images into our existing dataset was conducted with strategic care. Our goal was to enrich the dataset without introducing biases by applying manual labeling for all 10 classes, ensuring a balanced representation of each bird species. This augmented dataset promised a richer, more diverse training experience for our classification model.

Upholding quality standards: Quality assurance was pivotal in this endeavor. A rigorous review of the generated images was conducted, evaluating their realism and relevance for training purposes. Only those images that met our high standards of quality and authenticity were included in the augmented dataset.

In summary, the journey of training StyleGAN3 on our high-performance Tesla V100 GPUs, using PyTorch, and leveraging its capabilities for data augmentation was an exercise in precision, patience, and commitment to quality. This endeavor not only enhanced our dataset but also exemplified our dedication to adopting cutting-edge technology to advance the field of ecological image analysis.

3.3. The Proposed Deep Learning Model for Multi-class Classification

Algorithm 1 Deep learning model for multi-Class bird species classification.

Require: X, y Ensure: Model trained for bird species classification 1: if any of X, y is null or empty then 2: return error with message “Empty dataset or labels” 3: if data in X or y is not properly formatted then 4: return error with message “Invalid data format” 5: Initialize the model with input shape and number of classes 6: if model initialization fails then 7: return error with message “Model initialization failed” 8: Train the model using $X_{\text{train}}, y_{\text{train}}$ 9: if training fails then 10: return error with message “Model training failed” 11: Validate the model using $X_{\text{val}}, y_{\text{val}}$ 12: if validation accuracy is below threshold then 13: Adjust model parameters or training process 14: Test the model using $X_{\text{test}}, y_{\text{test}}$ 15: return model performance metrics (accuracy, precision, recall, F1 score)	▸ Dataset and labels
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To accurately classify bird species from the augmented dataset, we developed a deep learning model tailored for multi-class classification. This model was designed to harness the rich and complex features of the original and StyleGAN3-generated images.

We employed CNN architecture, renowned for its success in image recognition tasks. The CNN was constructed with multiple convolutional layers to extract hierarchical feature representations from the images, pooling layers to reduce dimensionality, and fully connected layers to perform the classification. The depth and width of the network were carefully calibrated to accommodate the complexity of the task, ensuring that the model could capture the fine-grained details necessary for distinguishing between the visually similar bird species.

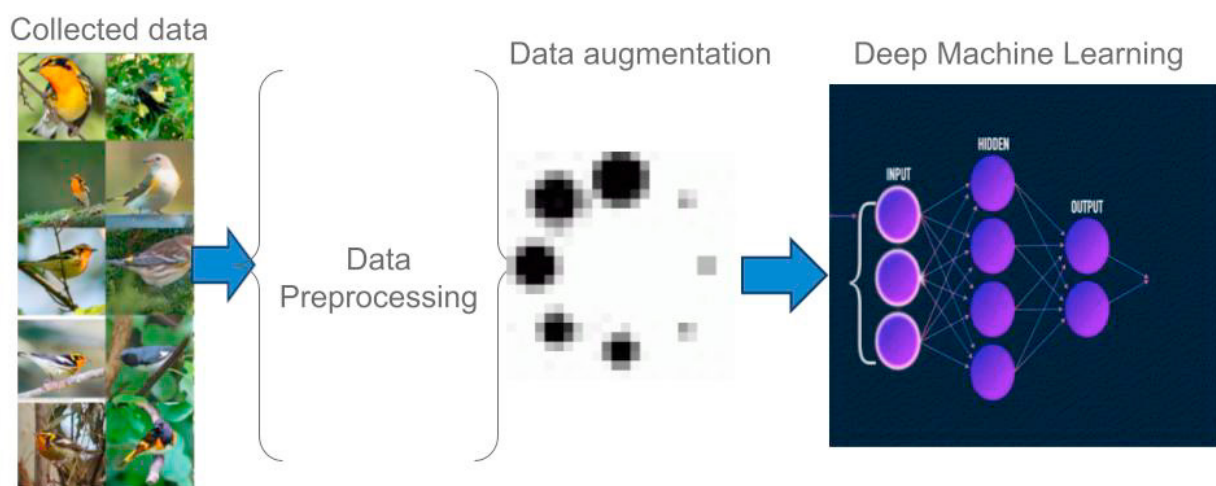


Fig. 2. Pipelining Methodology

Model Training: The dataset, now augmented with synthetic images from StyleGAN3, was divided into three subsets: 70% for training, 15% for validation, and 15% for testing. This division was crucial to evaluate the model's performance and mitigate overfitting accurately. We employed data shuffling and k-fold cross-validation to ensure the robustness and generalizability of the model.

Given its effectiveness in multi-class classification problems, we utilized cross-entropy loss as our objective function for the training process. The Adam optimizer was chosen for its adaptive learning rate capabilities, which allowed for more efficient convergence to the global minimum of the loss function. Regularization techniques, including dropout and L2 regularization, were applied to prevent overfitting.

Training was conducted over numerous epochs, with early stopping implemented based on the validation loss to prevent overtraining. The batch size was selected to optimize the computational efficiency while allowing for sufficient stochasticity during the gradient descent.

Performance evaluation metrics To assess the model's classification performance, we used several metrics: accuracy, precision, recall, and the F1 score. Accuracy provided a straightforward measure of overall performance, while precision and recall offered insight into the model's ability to correctly identify each class without misclassifying other classes. The F1 score, a harmonic mean of precision and recall, was instrumental in evaluating the model's performance on imbalanced datasets.

Hyperparameter tuning and model selection Hyperparameter tuning was conducted using a grid search approach, where a range of values for each hyperparameter was tested. The final model selection was based on the performance of the validation set, with the best-performing model being further evaluated on the test set to estimate its real-world performance.

Model interpretability To understand the decision-making process of the CNN, we employed visualization techniques such as feature map activation and saliency maps. These techniques allowed us to interpret which features of the bird images were most influential in the classification process, ensuring that the model's predictions were based on relevant visual features rather than spurious correlations.

In summary, the development and training of the CNN for multi-class bird species classification were executed meticulously, as shown in Algorithm 1, leveraging the enriched dataset to create an accurate and interpretable model. As shown in the pipelining methodology Fig. 2, this model is the culmination of our efforts to integrate advanced deep learning techniques with innovative data augmentation methods, setting a new benchmark for classification tasks in ecological research.

4. Results

This section of the paper outlines the manual data collection process and presents the results of our multi-classification model. It focuses on classifying labels with high similarity and demonstrates how using GANs for data augmentation has enhanced the model's performance with our meticulously collected dataset.



Fig. 3. Representative Images of the Ten Bird Species in the Dataset: Each image showcases distinctive features of a different species, illustrating the diversity and complexity in the dataset used for classification.

4.1. Data Collection

In this study, we utilize a meticulously assembled dataset of images from ten bird species, each represented by 950 manually selected images. These species are American Redstart, Black-throated Blue Warbler, Black-throated Green Warbler, Blackburnian Warbler, Chestnut-sided Warbler, Magnolia Warbler, Palm Warbler, Pine Warbler, Yellow Warbler, and Yellow-rumped Warbler. The collection process involved an extensive and careful manual search and selection from Google Images, ensuring a comprehensive and accurate representation of each species in their natural habitats.

The dataset features a balanced diversity of visual attributes, with each class corresponding to a specific bird species. It uses high-resolution, full-color images to capture the birds in various natural settings, introducing multiple background elements. This variety in backgrounds adds a layer of complexity to the classification task, necessitating sophisticated techniques to distinguish and classify each species accurately.

Providing detailed visual information crucial for identifying the subtle differences among these similar-looking bird species, this dataset serves as a rich resource for classification. A representative image from each species is included (see Fig. 3), offering a glimpse into the dataset's composition. These images highlight the variations in plumage coloration, bird postures, and environmental settings, emphasizing the classification challenges inherent in this diverse collection of avian imagery.

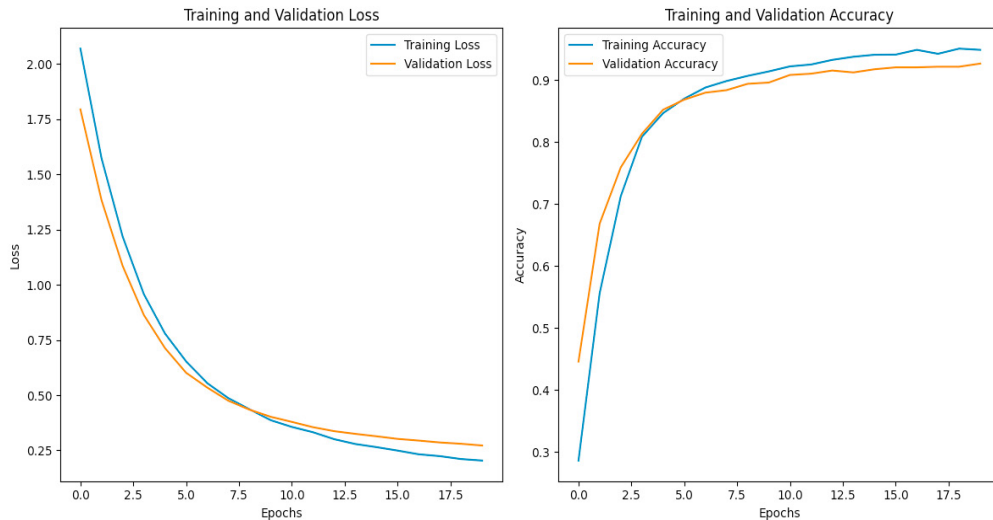


Fig. 4. Training and Validation Loss & Accuracy Curves with StyleGAN3-Augmented Data.

4.2. Model Performance with Data Augmentation

The application of GANs such as StyleGAN3 for data augmentation has shown to be highly beneficial for the classification performance of our model. As depicted in Fig. 4, the training and validation loss curves demonstrate a steady decline, indicative of effective model learning. The accuracy curves converge towards a high value, suggesting that the model generalizes well on unseen data.

Table 1 illustrates the aggregated classification metrics for the model trained on the StyleGAN3-augmented data, showing an overall accuracy of 93%.

Table 1. Aggregated Classification Metrics with StyleGAN3-Augmented Data

Metric	Score (%)
Accuracy	93
Macro Avg Precision	92
Macro Avg Recall	91
Macro Avg F1-Score	91
Weighted Avg	93

4.3. Model Performance without Data Augmentation

Conversely, when trained on the original dataset, the model's performance metrics, without augmentation, are provided in Table 2. The model shows good performance, yet there is a noticeable improvement compared to the results obtained with augmented data.

Table 2. Summary of Classification Metrics without Data Augmentation

Metric	Overall Score (%)
Accuracy	89
Macro Avg Precision	88
Macro Avg Recall	87
Macro Avg F1-Score	87
Weighted Avg	89

4.4. Comparative Analysis and Discussion

The comparison between the two sets of results in Table 1 and Table 2 clearly illustrates the benefits of incorporating StyleGAN3-generated images into the training process. The accuracy improvement by four percentage points is substantial for our fine-grained classification task. These results confirm our hypothesis that including StyleGAN3-augmented images can significantly enhance the performance of deep learning models on tasks involving the classification of visually similar entities, such as bird species, in our study.

This research has practical applications in biodiversity conservation, enabling more accurate monitoring and tracking of bird species. Additionally, the methodology can be adapted for use in agriculture for pest identification, in forestry for monitoring wildlife populations, and in environmental science for studying species diversity and ecosystem health. However, a limitation of this study is the reliance on synthetic data generated by GANs, which may not perfectly capture the ecological nuances of real-world bird species. Future research could focus on refining GAN models to better mimic these subtleties and validating the augmented datasets with field data.

5. Conclusion

This paper presents an innovative model for classifying labels with high similarity and subtle variations, addressing challenges posed by a limited and scarce dataset through meticulous manual data collection. To enhance the dataset's volume and diversity, we employed StyleGAN3-based Generative Adversarial Networks (GANs), significantly augmenting our manually collected data. This approach improved classification accuracy from 89% with the original dataset to 93% with the augmented dataset. This work exemplifies the synergy between human-driven data collection and advanced AI techniques, setting a new standard in machine learning applications for complex classification tasks. Lastly, the source code¹ and the dataset² are both made publicly available.

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¹ https://github.com/AliAlfatemi/ML_project

² https://github.com/AliAlfatemi/ML_project/tree/main/Dataset

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